

On-chip entangled photon source for robust and miniaturizable quantum communication transmitter

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Abstract

In the field of quantum-safe communication entanglement is a fundamental resource in many quantum communication and information processing protocols. For example, device-independent quantum key distribution can only be achieved through entanglement. It is therefore critical to implement high-performance, efficient entangled photon sources (EPSs), desirably on-chip. This would make EPSs a truly scalable mass-manufacturable technology. This work focuses on the experimental demonstration of entanglement generation in photonic integrated circuits (PICs), as well as its application for quantum-safe communication across free-space optical links, as envisioned for a future satellite-based quantum communication network.

Experimental Benchmark

The transmitter is based on Sagnac loops operating at 1550 nm. To prevent polarization-dependent losses from degrading entanglement fidelity, three chip copies were screened. Chip #3 was selected, as it provided balanced polarization beam splitter extinction ratios for TE and TM modes.

Characterized Metric	Unit	Chip #1	Chip #2	Chip #3
Linear Waveguide Parameters				
Average Propagation Loss*	dB/cm	1.22	0.98	—
Average Coupling Loss*	dB/facet	2.05	1.91	—
Polarization Performance				
PBS Extinction Ratio (TE)	dB	23.40	13.90	12.23
PBS Extinction Ratio (TM)	dB	3.40	6.57	12.30
PDM Extinction Ratio (Top Port)	dB	7.70	13.76	11.50
PDM Extinction Ratio (Bottom Port)	dB	—	25.00	30.10
Selection Status	—	Rejected	Rejected	SELECTED

Table 1. Chips characterization, tests results.

Testing followed a dual-stage pipeline to transition from classical mapping to quantum validation. Initially, classical stimulated Four-Wave Mixing (FWM) integrated a tunable seed and pump laser, to measure the loop bandwidth on an optical spectrum analyzer.

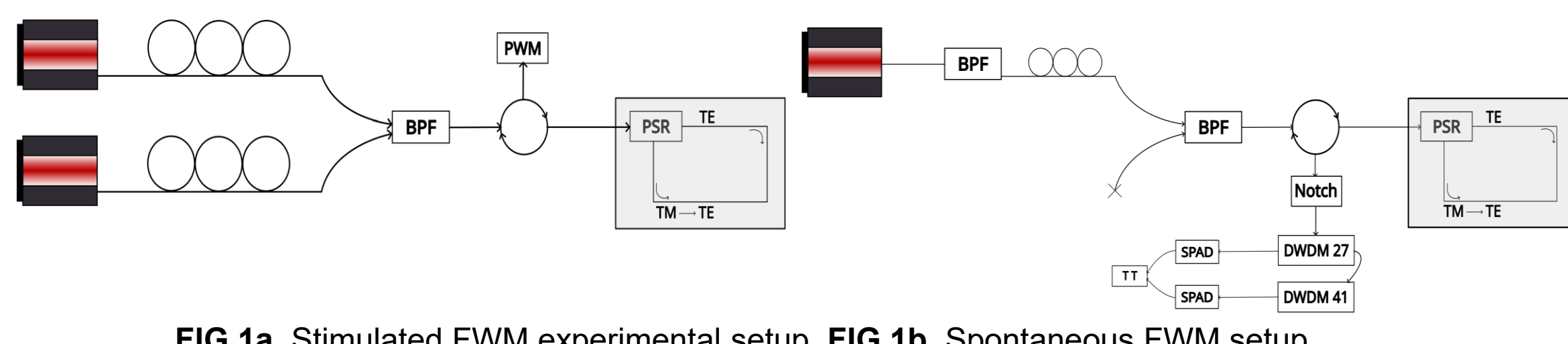


FIG 1a. Stimulated FWM experimental setup. FIG 1b. Spontaneous FWM setup.

Finally, the seed was disconnected to trigger quantum spontaneous FWM inside the Sagnac loop. To isolate the generated photon pairs from the pump background, we integrate an output filter cascade combining a notch filter and dense wavelength-division multiplexers; this way we suppress pump photons, routing the clean telecom signal and idler photons directly to the single-photon detectors.

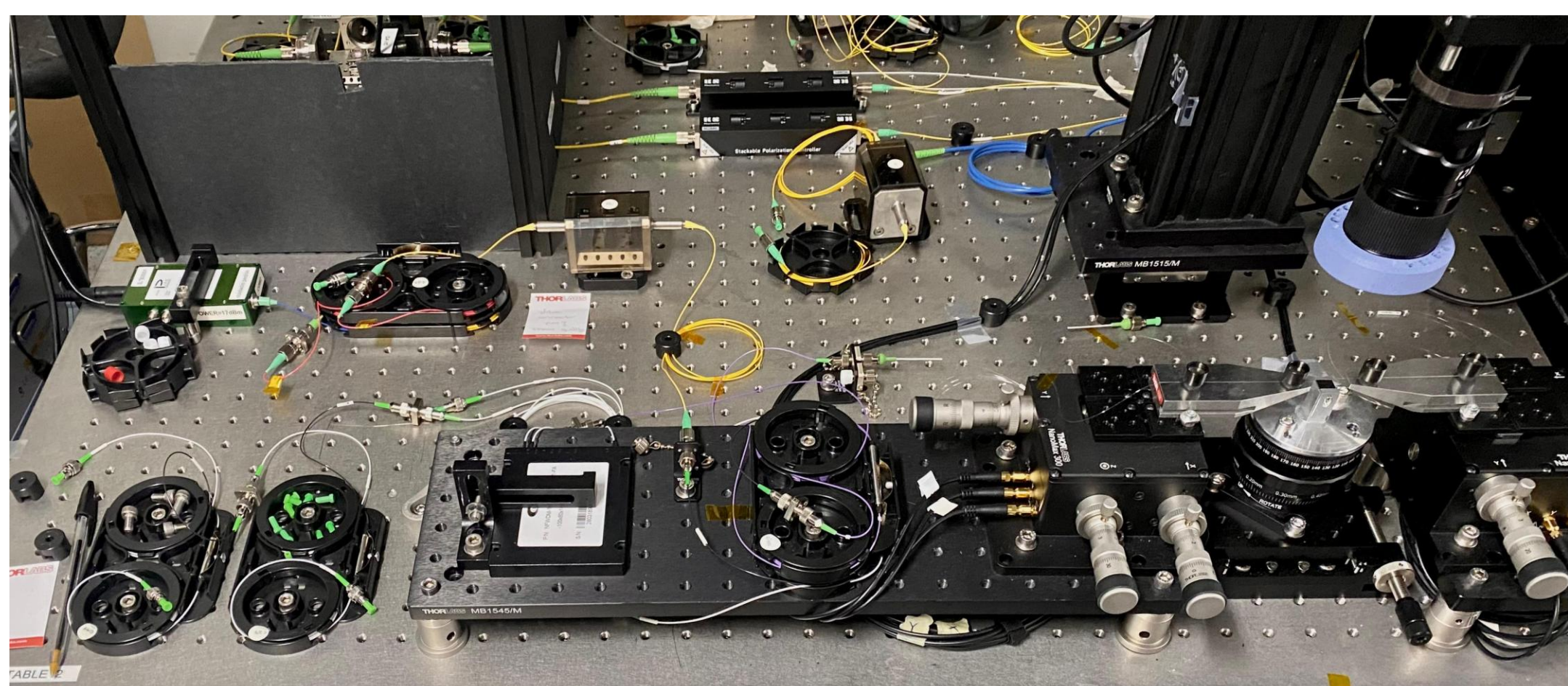


FIG 2. Laboratory implementation of setup FIG 1b.

Conclusion

We show that the presented on-chip entangled photon source shows high potential to be a high-fidelity candidate for future satellite-based quantum communication networks. Chip #3 was validated for its structural symmetry, yielding balanced PBS ER for TE (122.23 dB) and TM (12.30 dB) modes, alongside a 30.10 dB multiplexer isolation to mitigate polarization-dependent losses. Classical FWM isolated the superior waveguides, such that when transitioning to the quantum regime, spontaneous FWM generated correlated single-photon pairs.

Building upon this, future steps will focus on transitioning this integrated transmitter in a hardware-in-the-loop free-space link emulation. Using satellite trajectory data, we will calculate time-resolved atmospheric parameters, including the coherence width and Greenwood frequency, to simulate high-frequency turbulence fading. By feeding these loss profiles into a digital variable optical attenuator (DVOA), we will dynamically subject our quantum states to real-world atmospheric degradation. This emulation framework will allow to definitively benchmark the source's entanglement fidelity under realistic satellite downlink conditions.

References

[1] Xin Hua et al, A bright polarization-entangled photon pair source based on silicon photonics integration. APL Photonics 1 June 2025; 10 (6): 066109.

[2] Meritxell Cabrejo-Ponce et al 2022 Quantum Sci. Technol. 7 045022

Results

Due to its high polarization symmetry, Chip #3 was chosen as the optimal candidate, yielding nearly identical PBS ER for TE and TM. This is vital to mitigate polarization-dependent losses during satellite tracking.

Classical FWM sweeps mapped the performance of the six on-chip Sagnac loops. The raw sweeps reveal a sinusoidal Fabry-Perot ripples from facet reflections, and efficiency computation shows the superior performance of the $0.55\mu\text{m}$ width loop: wvg_2 and wvg_4 . Whilst wvg_6 showed a flatter efficiency bandwidth, the difference between TE and TM was apparent. Thus, wvg_2 and wvg_4 have been chosen as candidates for the quantum setup.

Moreover, the tracking of the clean spectral evolution confirms Continuous, stable photon generation across the experimental sweep window: 1480-1625 nm.

Quantum Realization

Transitioning to the quantum regime confirms the successful generation of correlated single-photon pairs via spontaneous FWM: the time-tagger reveals a distinct, narrow coincidence peak. This milestone confirms that the selected on-chip Sagnac loop and setup produce and isolates correlated photon pairs. However, further measurements are required to quantify the state Fidelity and certify true non-local entanglement.

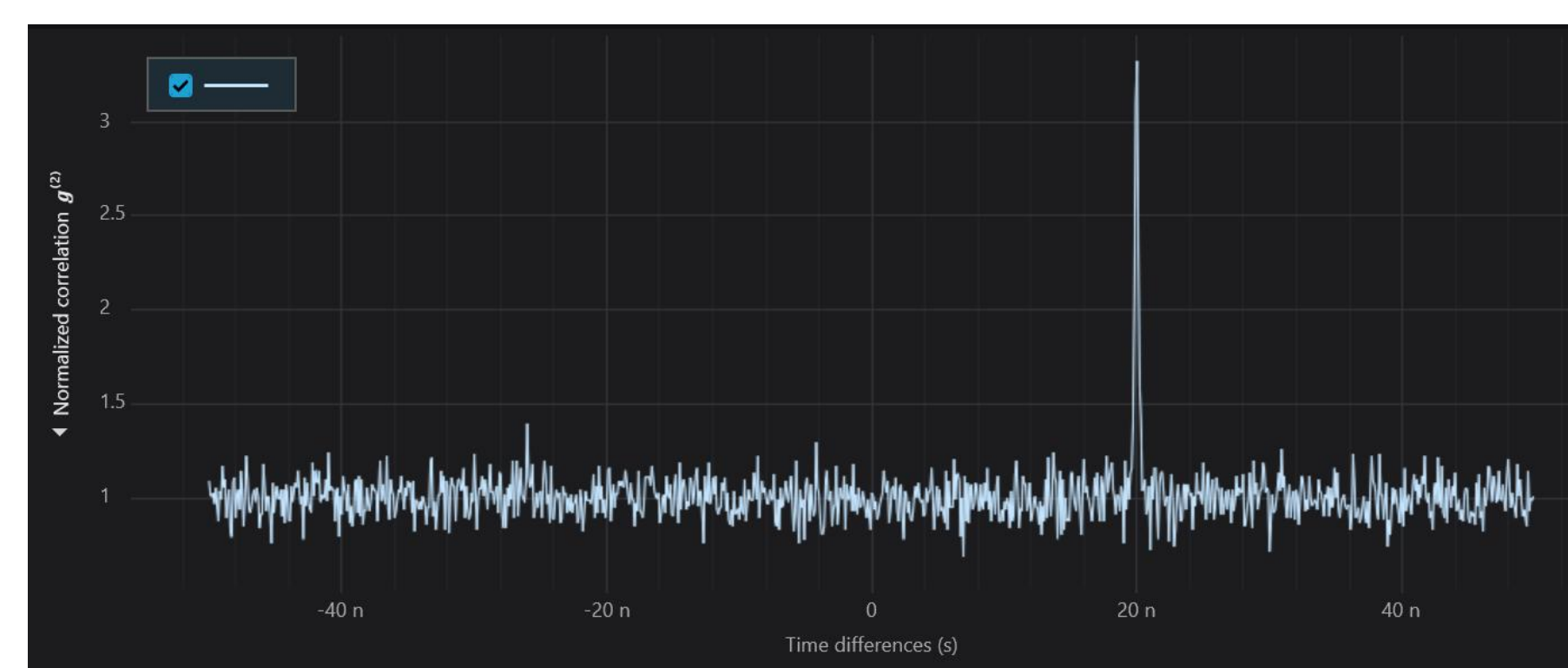


FIG 6. Time-tagger plot showing entangled photon coincidences

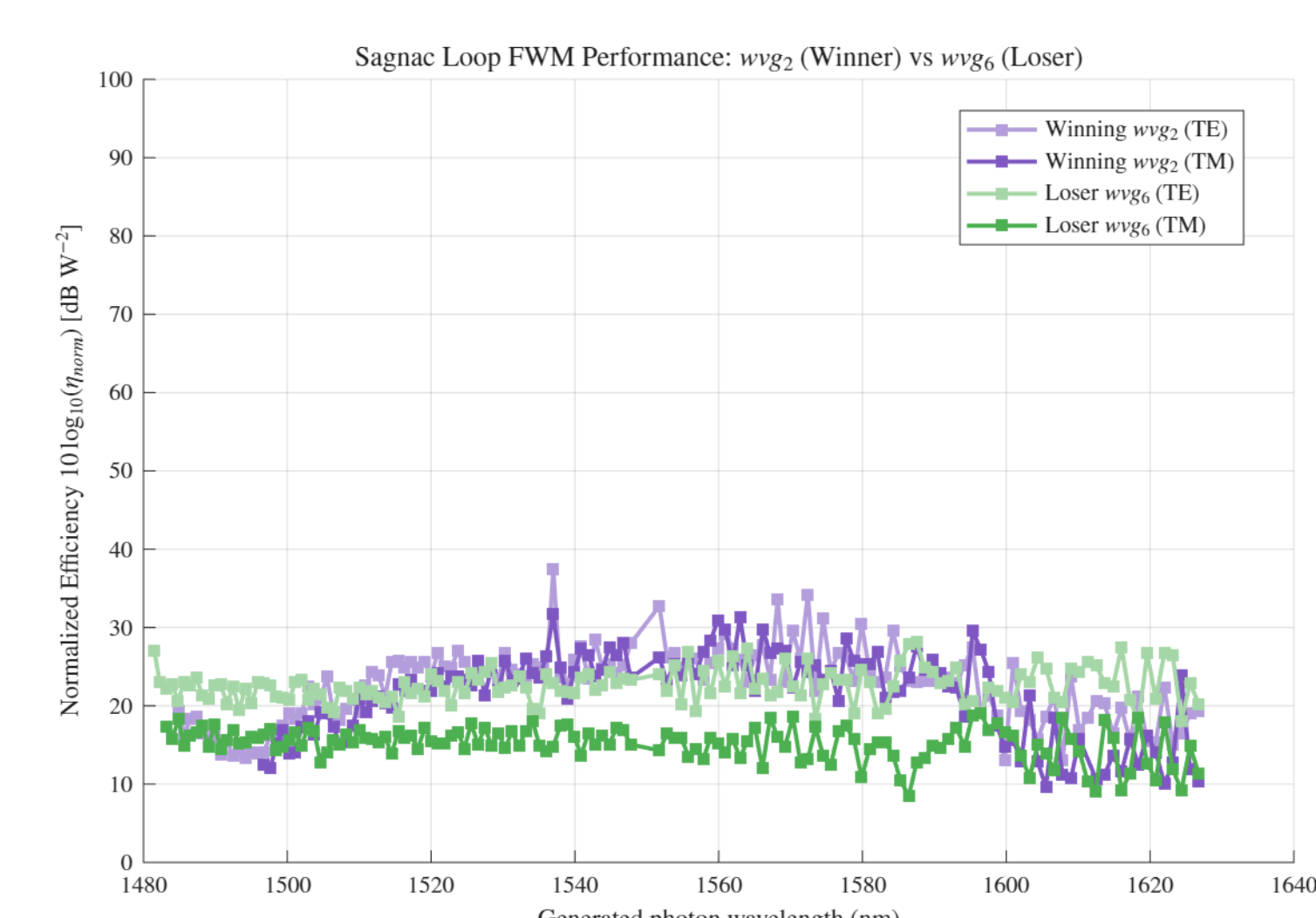


FIG 3. Conversion efficiency

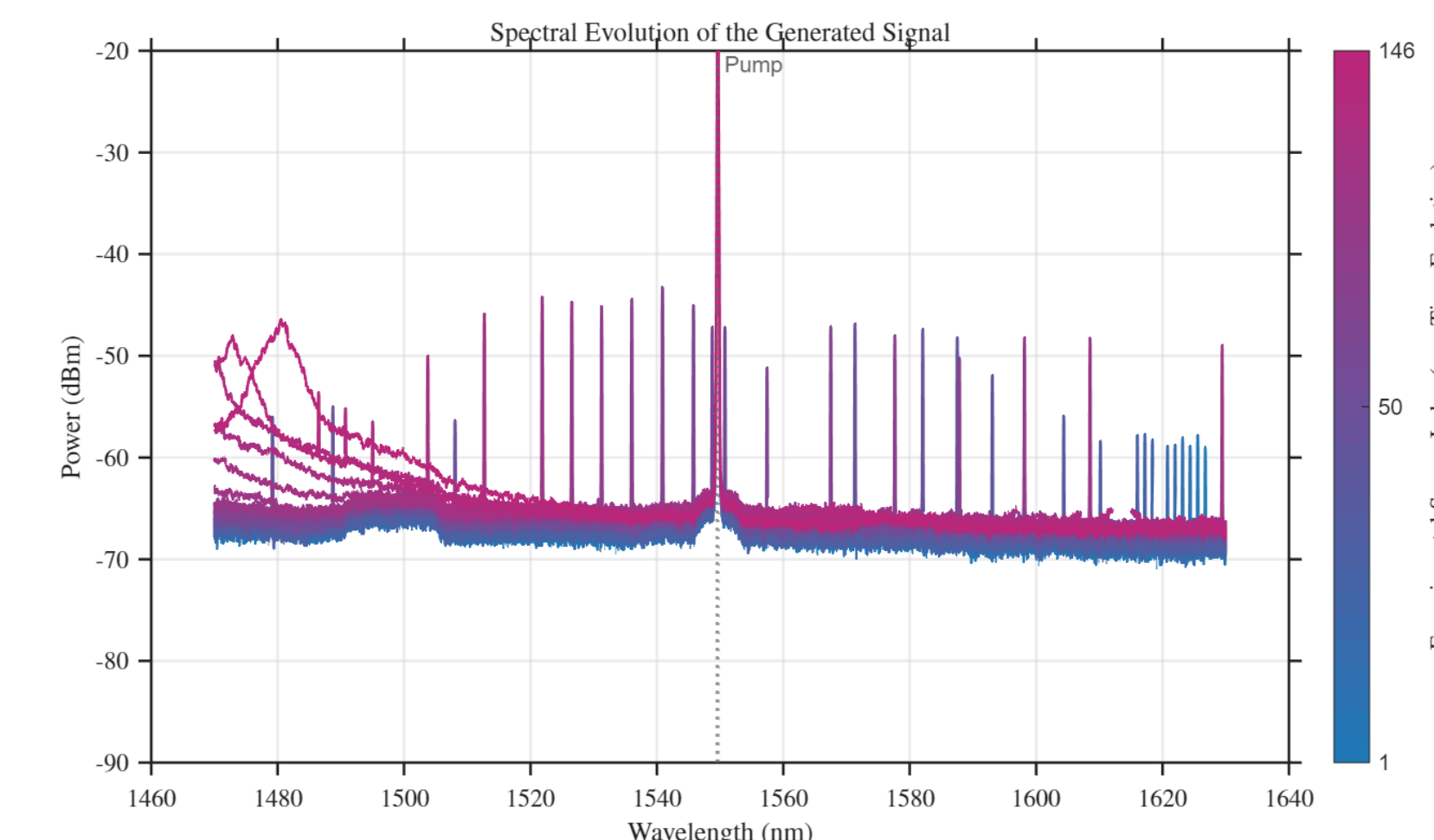


FIG 4. Generated signal spectral evolution

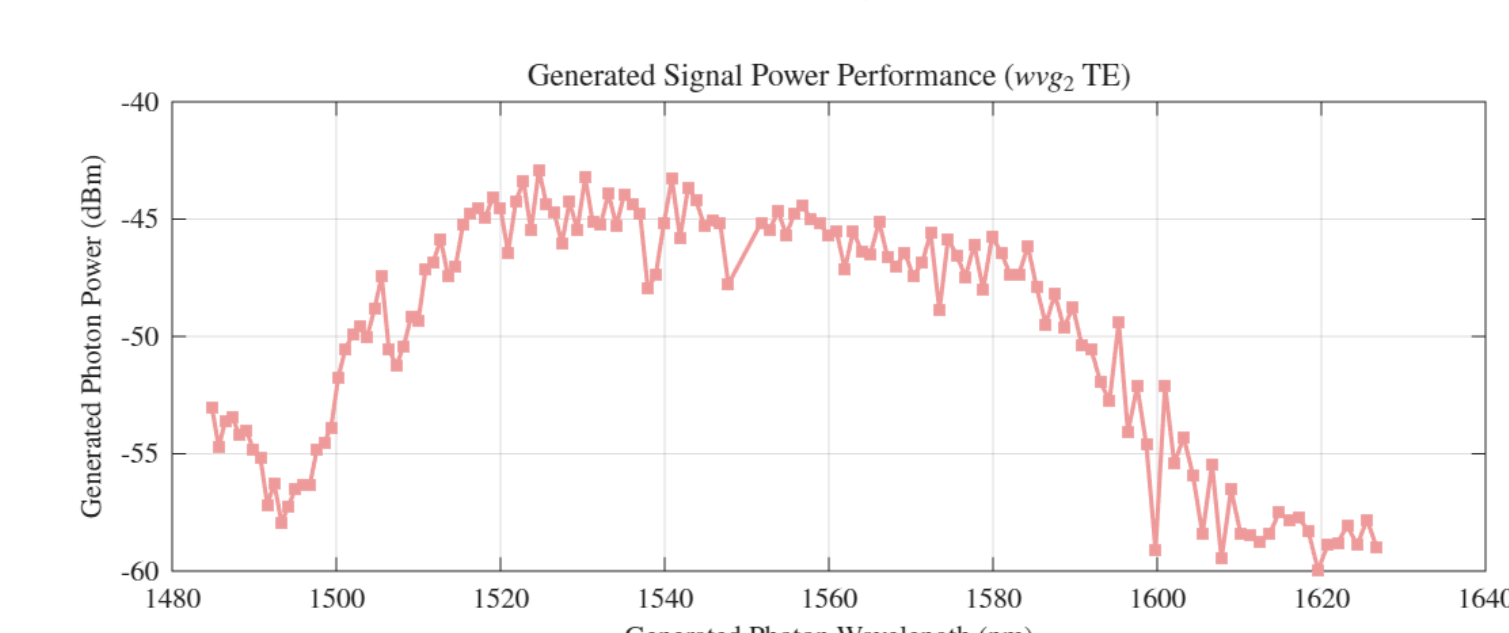
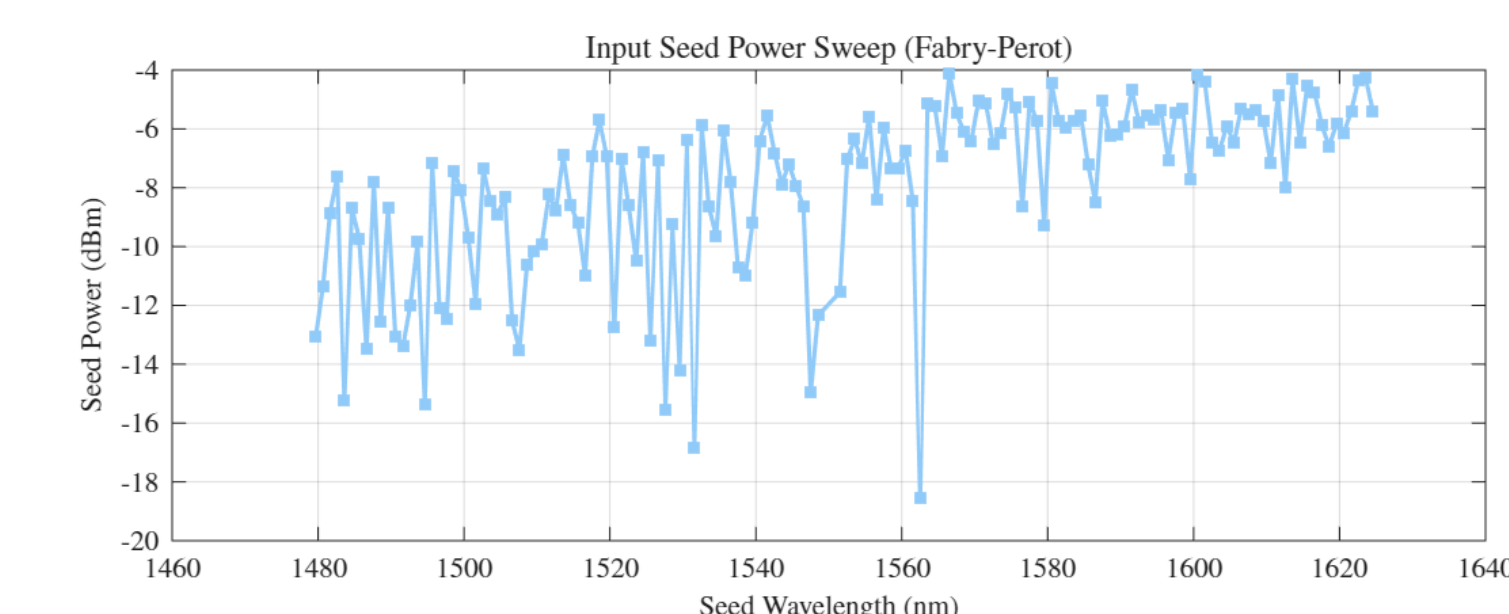


FIG 5a. Seed power sweep. FIG 5b. Generated signal power sweep

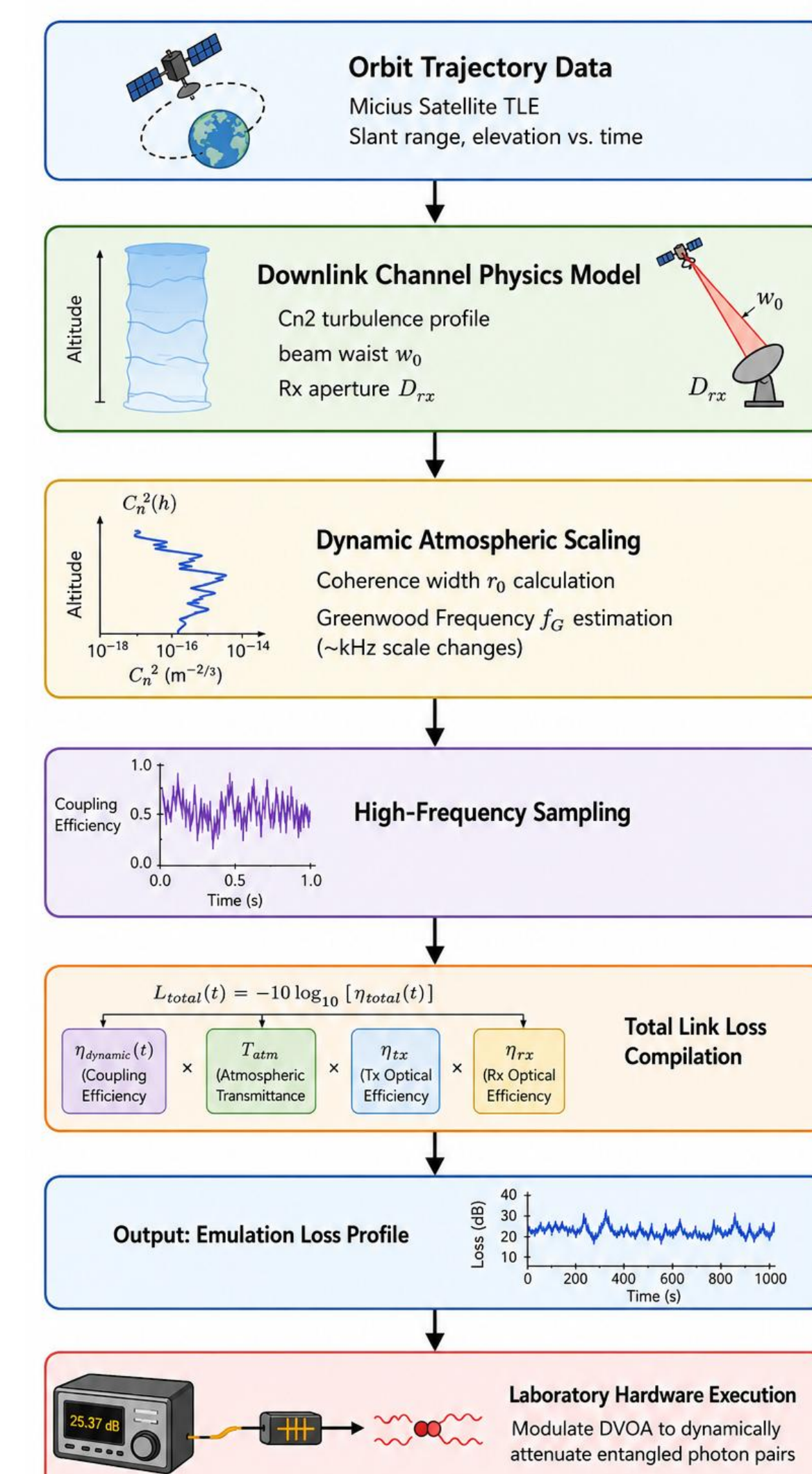


FIG 7. Free-space link emulation workflow.